Topic 5 — Sound and Wave Motion

MANDATORY SYLLABUS SCOPE: Covers all dimensions of Wave Motion (Mechanical, Transverse, Longitudinal, Surface water waves, Electromagnetic waves, EM Spectrum, Radio wave propagation, Terrestrial radiation, Albedo) and Acoustics (Speed of sound across media, Factors affecting speed, Pitch/Loudness/Timbre, Echo & Reverberation, Infrasound, Ultrasound, SONAR, Noise pollution decibel standards, Diatonic scale, Doppler effect, Mach number). Fully synchronized with Ghatna Chakra Physics, NCERT Class 9–11, and latest 2021–2025 State PCS PSC papers.

Current Affairs (2024–2026 Focus)

1. **UNEP 'Frontiers Report' on Noise Pollution (Moradabad Spotlight):** - In the UNEP report *Noise, Blazes and Mismatches*, **Moradabad (Uttar Pradesh)** was documented recording peak noise levels of $114\ \text{dB}$, ranking as the second noisiest city in the world after Dhaka, Bangladesh ($119\ \text{dB}$), and ahead of Islamabad ($105\ \text{dB}$). - **UPPCS Direct Question Fact:** WHO recommends ambient residential noise levels below $45\ \text{dB}$. Any continuous exposure above $80\ \text{dB}$ constitutes hazardous noise pollution causing permanent auditory damage, hypertension, and sleep disorders.
2. **Insolation vs Terrestrial Radiation (IAS Pre 2024 / UPPCS 2022 Fact):** - Earth receives incoming solar energy primarily in **shortwave radiation** (visible and near UV/IR). The atmosphere is largely transparent to incoming shortwave solar radiation (14% absorbed directly by air, 51% by the Earth's surface, 35% reflected by clouds/aerosols/albedo). - Earth heated by insolation re-radiates energy back as **longwave terrestrial radiation (infrared)**. Greenhouse gases ($\text{CO}_2, \text{CH}_4, \text{H}_2\text{O}$ vapour) strongly absorb this longwave radiation. Thus, **the Earth's atmosphere is heated primarily from below by longwave terrestrial radiation**, NOT directly by incoming sunlight!
3. **Albedo Measurements in Climate Science (UPPCS Pre Fact):** - **Fresh snow cover** exhibits the highest albedo on Earth ($80\% - 95\%$ reflecting power), causing skiers to suffer UV burns from reflected radiation. Open water exhibits the lowest albedo ($\sim 3\% - 5\%$).
4. **Telecommunication Frequency Bands & Space Waves (RAS Pre / UPPCS Fact):** - **Ultra High Frequency (UHF: $300\ \text{MHz} - 3\ \text{GHz}$)** cannot follow the curvature of Earth (ground waves) nor be reflected by the ionosphere (sky waves). They propagate via **space waves (line-of-sight propagation)**, which powers mobile telephony, Wi-Fi, and UHF television broadcasting.
5. **Acoustic Bathymetry & Deep Sea Submersibles (Samudrayaan / MATSYA 6000 Fact):** - Radio waves and light cannot penetrate deep seawater due to massive electrical conductivity and absorption. Only **acoustic pulses (ultrasonic SONAR)** can navigate, map seabed bathymetry, and communicate under deep oceanic trenches.

Detailed In-Depth Core Theory

5.1 Wave Motion: Mechanical vs Electromagnetic

1. **Mechanical Waves:** - Require a continuous elastic medium endowed with mass (inertia) and elasticity (restoring capability). - **Transverse Waves:** Particle motion is strictly orthogonal to wave propagation. Propagates via **shear stress** (crests and troughs). Can only propagate through media that possess shear modulus (solids, and specialized orbital motion on liquid surfaces). They **cannot propagate through the bulk interior of liquids or gases**. - **Longitudinal Waves:** Particle displacement is parallel to wave propagation direction, creating regions of compression (high density, high pressure) and rarefaction (low density, low pressure). Propagate via **bulk modulus**; can exist in solids, liquids, and gases.
2. **Electromagnetic Waves:** - Originate from accelerated electric charges. Maxwell showed that time-varying electric fields produce magnetic fields, and vice versa. - Travel through vacuum at $c = 1/\sqrt{\mu_0 \varepsilon_0} \approx 3.0 \times 10^8\ \mathrm{m/s}$. - Transverse in nature: $\vec{E} \perp \vec{B} \perp \vec{v}$.

5.2 Sound Propagation and Speed Determinants

1. **Newton-Laplace Formula:** - Laplace corrected Newton's isothermal assumption by proving that sound compressions/rarefactions occur adiabatically:

$$v = \sqrt{\frac{\gamma P}{\rho}} = \sqrt{\frac{\gamma R T}{M}}$$

   - Where $\gamma = C_p/C_v$ (adiabatic index, ≈ 1.4 for air), P is pressure, ρ is density, T is absolute temperature, and M is molecular mass.
2. **Determinants of Speed of Sound:** - **Effect of Temperature:** $v \propto \sqrt{T}$. Speed increases by approximately $0.61\ \mathrm{m/s}$ for every $1^\circ\mathrm{C}$ temperature rise in air. - **Effect of Humidity:** Moist air contains water vapour ($M = 18$), which is lighter than dry air ($M \approx 29$). Thus, humid air is less dense than dry air, and the speed of sound is **greater in humid air**. - **Effect of Pressure:** Pressure changes at constant temperature cause density to change in identical proportion ($P/\rho = \text{constant}$). Hence, **speed of sound is independent of pressure**. - **Effect of Medium State:** Rigidity/Elasticity determines speed: $v_{\text{solid}} > v_{\text{liquid}} > v_{\text{gas}}$.

5.3 Characteristics of Sound: Pitch, Loudness and Timbre

1. **Pitch (Frequency):** - Sensation of the frequency of sound. High pitch corresponds to high frequency and shorter wavelength (shrill sound of female voice, mosquito whine). Low pitch corresponds to low frequency (deep roar of lion).
2. **Loudness (Amplitude):** - Objective intensity $I = P/A = \frac{1}{2}\rho v \omega^2 A^2$ (proportional to square of amplitude). Subjective sensation is loudness: $L = 10 \log_{10}(I/I_0)$ in decibels (dB).
3. **Quality / Timbre:** - Characteristic that enables distinction between two musical notes having the exact same pitch and loudness, caused by the number and relative intensities of **harmonics (overtones)**. - Phonemes are the basic structural sound units in linguistics.

5.4 Reflection, Echo and Reverberation

1. **Echo Physics:** - Persistence of hearing is $0.1\ \text{s}$. The reflected wave must return after $\geq 0.1\ \text{s}$. - Distance: $2d = v \cdot t \implies d = \frac{v \times 0.1}{2} = \frac{344 \times 0.1}{2} = 17.2\ \text{m}$.
2. **Reverberation:** - Prolongation of sound due to repetitive reflections in enclosed spaces. Controlled using Sabine's formula $T = 0.161V/A$ by adding absorbent acoustic tiles, heavy drapes, and upholstered seating.

5.5 Infrasound, Ultrasound and Technological Applications

1. **Infrasound ($< 20\ \text{Hz}$):** Generated before earthquake P-waves; perceived by elephants and whales for long-distance communication; stethoscope records cardiac infrasonic vibrations.
2. **Ultrasound ($> 20\ \text{kHz}$):** - **Echolocation:** Bats emit ultrasonic pulses ($20 - 100\ \text{kHz}$) and capture returning echoes to navigate and hunt in total darkness. - **Medical Imaging (Sonography & Echocardiography):** High-frequency acoustic pulses reflect from tissue interfaces without ionizing tissue damage (unlike X-rays). - **Lithotripsy:** Focused shockwaves fragment renal calculi. - **SONAR (Sound Navigation and Ranging):** Emits acoustic ultrasound pulses into seawater; calculates target distance via $d = v \cdot t/2$.

5.6 Seismic Waves, Radio Propagation & Atmospheric Energetics

1. **Earthquake Waves:** P-waves (compressional/longitudinal, fastest, all states) vs S-waves (shear/transverse, slower, solids only).
2. **Radio Communication Modes:** - **Ground Waves:** Follow Earth curvature; attenuated at high frequencies ($< 2\ \text{MHz}$). - **Sky Waves:** Bounded by total internal reflection from the Ionosphere layer ($2 - 30\ \text{MHz}$). - **Space Waves:** Line of sight; UHF ($300\ \text{MHz} - 3\ \text{GHz}$) passes through the ionosphere into space; used for satellite and line-of-sight communications.
3. **Atmospheric Heat Engine:** - Earth is heated from below: Sun emits shortwave radiation $\rightarrow$ Earth absorbs $\rightarrow$ Earth emits longwave infrared $\rightarrow$ Absorbed by greenhouse gases.

Bilingual Terminology -- Sound and Wave Motion

English Term	Hindi Term	Key Anchor
Sound (ध्वनि)	ध्वनि	Mechanical longitudinal wave; needs medium; cannot travel in vacuum
Frequency (आवृत्ति)	आवृत्ति	Number of vibrations per second; SI unit = Hz
Wavelength (तरंगदैर्ध्य)	तरंगदैर्ध्य	Distance between two consecutive crests/troughs; SI unit = metre
Amplitude (आयाम)	आयाम	Maximum displacement from mean position; determines loudness
Pitch (तारत्व)	तारत्व	Highness/lowness of sound; depends on frequency
Loudness (प्रबलता)	प्रबलता	Subjective sensation; depends on amplitude and intensity
Infrasound (अवश्रव्य ध्वनि)	अवश्रव्य ध्वनि	Frequency < 20 Hz; humans cannot hear; elephants, whales use it

English Term	Hindi Term	Key Anchor
Ultrasound (पराश्रव्य ध्वनि)	पराश्रव्य ध्वनि	Frequency > 20,000 Hz; bats, dolphins; used in SONAR, medical imaging
Echo (प्रतिध्वनि)	प्रतिध्वनि	Reflection of sound; minimum distance for echo = 17 m (at 20 degC)
Reverberation (अनुनाद)	प्रतिध्वनि लम्बी / अनुगूँज	Persistence of sound due to multiple reflections in enclosed space
Resonance (अनुनाद)	अनुनाद	When applied frequency = natural frequency; amplitude becomes large
Doppler Effect (डॉपलर प्रभाव)	डॉपलर प्रभाव	Change in apparent frequency due to relative motion between source and observer
Decibel (dB) (डेसिबल)	डेसिबल	Logarithmic unit of sound intensity
SONAR (सोनार)	सोनार	Sound Navigation and Ranging; uses ultrasound to detect underwater objects
Noise Pollution (ध्वनि प्रदूषण)	ध्वनि प्रदूषण	Unwanted sound causing health problems; >85 dB prolonged = hearing damage

One-Liner Revision -- Sound and Wave Motion

#	Fast Fact
1	Sound is a **mechanical longitudinal** wave (particles vibrate parallel to wave direction)
2	Sound needs a **medium** to travel; cannot travel in vacuum
3	Speed of sound: 343 m/s in air (20 degC); 1500 m/s in water; 5100 m/s in steel
4	Speed of sound increases with temperature and humidity
5	Sound travels fastest in **solids**, slowest in **gases**
6	Human audible range = **20 Hz to 20,000 Hz**
7	Infrasound (< 20 Hz) = elephants, whales, earthquakes; long-range communication
8	Ultrasound (> 20,000 Hz) = bats (echolocation), dolphins; SONAR; medical USG
9	SONAR uses ultrasound to detect submarines, fish, ocean depth
10	Medical ultrasonography = uses 1-20 MHz ultrasound to image internal organs
11	Echo = minimum distance = **17 metres** (at 20 degC); time gap > 0.1 sec
12	Reverberation = multiple reflections; undesirable in auditoriums; curtains absorb
13	Resonance = when forced frequency = natural frequency; Tacoma Narrows Bridge collapse (1940)
14	Doppler Effect: source approaching = frequency increases (blue shift); receding = decreases (red shift)

#	Fast Fact
15	Doppler Effect used in: radar speed guns, echocardiography, astronomy (red shift)
16	Mach number = object speed / speed of sound; Mach 1 = speed of sound
17	Sonic boom = shock wave when object crosses Mach 1 (supersonic)
18	dB levels: Whisper = 20 dB; Conversation = 60 dB; Rock concert = 110-120 dB
19	Safe noise level = <85 dB (prolonged); >140 dB = pain threshold
20	Noise pollution limit (India): day = 55 dB (residential), night = 45 dB
21	Beats = alternating loud/soft sound when two slightly different frequencies combine
22	Beat frequency =
23	Musical instruments: String (sitar), Wind (flute, trumpet), Percussion (tabla)
24	Stethoscope = uses sound conduction (not amplification) to hear heartbeats
25	Thunder heard after lightning = because light (3x10^8 m/s) >> sound (343 m/s)
26	Sound wavelength (audible range): 1.7 cm (20 kHz) to 17 m (20 Hz)
27	Intensity of sound: I = P/(4*pi*r^2); decreases as 1/r^2 from point source
28	Transverse waves: light, EM waves; Longitudinal waves: sound, seismic P-waves
29	Seismic P-waves = longitudinal; S-waves = transverse (can't pass through liquid outer core)
30	Infrasound from earthquakes detected by animals before humans = early warning